

BRANDVLEI, South Africa

WMO: 685230

Lat: 30.4647S

Lon: 20.4783E

Elev: 3025

StdP: 13.16

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.4	33.1	3.8	7.5	55.1	7.1	8.8	52.5	29.5	34.5	21.6	63.9	2.7	200	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.6	100.4	63.0	97.8	62.3	95.3	61.4	68.3	83.2	67.0	83.0	65.6	84.0	10.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	100.4	74.4	61.8	92.5	74.8	59.0	83.5	74.6	34.4	82.5	33.3	82.1	32.1	83.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	19.0	17.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.4	80.3	79.2	75.7	67.8	59.2	52.0	51.5	54.2	60.8	67.8	72.5	76.8
	DBStd	12.20	5.70	5.77	6.12	7.10	6.78	5.69	6.73	6.85	7.00	8.12	7.50	5.95
	HDD50	146	0	0	0	0	8	40	63	31	4	0	0	0
	HDD65	1675	0	1	6	56	202	390	420	344	165	67	23	1
	CDD50	6138	941	818	796	535	294	100	110	160	327	551	675	831
	CDD65	2193	476	399	337	140	23	1	2	9	38	153	248	367
	CDH74	26054	5966	4571	3618	1319	336	28	39	186	584	1872	2992	4543
	CDH80	13012	3348	2433	1757	465	70	1	0	41	193	860	1439	2405
Wind	WSAvg	7.9	9.3	8.5	7.4	6.6	6.3	6.6	7.1	7.1	7.6	8.8	9.5	9.7
Precipitation	PrecAvg	5.4	0.6	0.7	1.0	0.8	0.3	0.3	0.3	0.1	0.2	0.5	0.4	0.3
	PrecMax	14.8	3.0	3.6	4.6	5.4	2.3	1.4	3.0	1.1	1.3	3.3	2.8	2.0
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.3	0.8	0.9	1.1	1.1	0.4	0.4	0.6	0.2	0.3	0.8	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.9	102.1	99.0	92.7	85.9	77.8	78.2	85.1	90.3	98.0	99.5	102.0
		MCWB	64.2	65.0	62.9	61.6	57.7	54.1	53.2	55.1	57.4	61.5	60.7	62.0
	2%	DB	100.8	98.5	95.7	88.5	81.5	72.8	74.2	79.3	85.3	93.3	95.2	98.3
		MCWB	63.1	64.0	62.2	60.0	56.1	51.7	51.6	53.1	55.6	59.3	59.0	61.6
	5%	DB	97.7	95.6	92.8	85.2	77.6	69.1	70.5	74.5	80.7	89.4	91.8	95.4
		MCWB	62.5	62.7	61.3	59.0	54.6	50.1	49.8	50.6	53.3	57.8	58.5	61.0
	10%	DB	94.5	92.6	89.6	81.5	73.4	64.5	66.1	69.5	75.7	84.8	88.4	92.1
		MCWB	61.5	61.6	60.5	57.8	53.5	48.9	48.4	49.4	51.1	55.9	57.2	60.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	70.2	68.7	64.9	60.9	55.6	54.6	58.3	60.2	63.8	64.7	67.6
		MCDB	84.7	87.6	81.5	80.5	76.8	71.5	73.6	71.1	86.6	90.4	87.7	87.7
	2%	WB	68.0	68.6	66.3	62.7	58.6	53.4	52.9	55.3	56.3	61.8	62.7	65.2
		MCDB	82.1	84.1	83.1	80.8	72.7	67.7	71.1	72.3	80.2	85.9	84.5	86.1
	5%	WB	66.5	67.3	65.0	61.3	56.9	51.8	51.1	52.7	54.3	59.8	60.3	63.6
		MCDB	84.4	82.0	83.0	79.3	73.2	64.4	67.7	71.7	77.1	83.3	85.4	86.1
	10%	WB	64.9	66.0	63.3	59.9	55.0	50.2	49.1	50.1	52.3	57.4	58.5	62.0
		MCDB	87.0	81.8	83.1	77.3	71.5	61.7	63.6	67.3	73.3	80.6	83.6	85.9
Mean Daily Temperature Range	5% DB	MDBR	26.6	25.5	26.4	24.3	25.6	24.6	26.7	28.6	27.3	27.5	27.4	28.2
		MCDBR	29.7	28.3	28.9	27.9	30.6	31.4	32.2	35.0	33.7	32.6	31.1	31.4
		MCWBR	9.8	9.8	10.3	10.9	13.6	15.6	16.3	16.6	15.7	12.3	11.9	11.0
	5% WB	MCDBR	23.0	21.2	22.9	23.5	25.7	25.2	27.9	29.6	30.4	29.9	28.4	26.8
		MCWBR	8.0	8.7	9.4	9.5	11.9	13.4	14.4	14.2	14.5	11.7	12.5	10.4
Clear-Sky Solar Irradiance	taub	0.309	0.319	0.311	0.292	0.273	0.260	0.263	0.292	0.307	0.318	0.295	0.299	
	taud	2.558	2.536	2.552	2.603	2.642	2.641	2.616	2.491	2.415	2.420	2.556	2.570	
	Ebn at Noon	327	318	311	301	294	291	294	296	306	315	330	331	
	Edh at Noon	34	34	31	27	23	22	23	30	35	38	34	34	
All-Sky Solar Radiation	RadAvg	2706	2444	2068	1595	1247	1077	1179	1476	1924	2337	2665	2830	
	RadStd	94	96	68	67	57	47	44	59	70	71	95	114	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.87	N/A	N/A	N/A	N/A

Nomenclature: See separate page